BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 155 · Paper 1: Software, Impacts and Legislation

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does encryption protect transmitted data?

Q2. What does FROM name in SQL?
A. The output columns
B. The filter condition
C. The source table
D. The sort order

Q3. “A syntax error is any incorrect answer produced by syntactically valid code.” Correct this. Explain why.

Q4. A program repeats the same price calculation in several places. “Copying long code blocks always makes maintenance easier.” Correct this. How would a subprogram help?

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I Do · Read new knowledge

Key vocabulary

Operating-system services — Manage memory, files, processes and peripherals.

Utility tools — Encryption, compression and HDD defragmentation serve different needs.

Societal impacts — Evaluate benefit and risk for identified stakeholders.

Legal framework — Match misuse, copyright and personal-data concerns to the appropriate legislation/rules.

New learning

Operating-system services: Manage memory, files, processes and peripherals.

Utility tools: Encryption, compression and HDD defragmentation serve different needs.

Societal impacts: Evaluate benefit and risk for identified stakeholders.

Legal framework: Match misuse, copyright and personal-data concerns to the appropriate legislation/rules.

Worked example

Evaluate a school cloud backup service in terms of benefits and risks.

1. Off-site backups can support data recovery.

2. Internet dependency and availability can limit access.

3. Student records need protected access and appropriate lawful handling.

4. Justify using secure, tested backup procedures.

We Do · Practise with guidance

Guided 1

A staff member uploads copyrighted media without permission. Identify the issue.

Worked method: Conclude possible copyright infringement.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the Boolean expression that rejects collecting data without a purpose.

```
hasPurpose = false
if ... then
    print("Collect data")
else
    print("Do not collect")
endif
```

write

Question 1

Give one benefit of OS multitasking. [2 marks]

Question 2

What is the main purpose of a compression utility? [2 marks]

Question 3

Identify relevant law for accessing another user's account without permission. [1 mark]

Exit Ticket · Answer before checking

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Operating systems can manage more than one running process.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Encryption changes plaintext into ciphertext. Someone who intercepts it cannot easily read it without the key.

Q2. C — The source table

Q3. A syntax error breaks the language's rules. The affected code cannot be translated or run correctly.

Q4. Put shared code in one subprogram. It can be reused, tested and changed in one place.

I Do / We Do pages · Guided practice

Q1. Conclude possible copyright infringement.

You Do pages · Code First

```
hasPurpose = false
if hasPurpose then
    print("Collect data")
else
    print("Do not collect")
endif
```

Expected result: Do not collect

You Do pages · Independent questions

Q1. Allows multiple applications/processes to make progress through scheduling.

Q2. Reduce the size of stored/transmitted data.

Q3. Computer Misuse Act 1990.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: